

Proposal: Distributional Neural Fictitious Self-Play for Imperfect-Information Games

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Project type: Option 4: Your own novel research!

Description This project aims to conduct novel research, building upon the reinforcement learning method proposed by Heinrich and Silver [2]. It will investigate Neural Fictitious Self-Play (NFSP) [2], which is a deep multi-agent framework that learn to approximate Nash equilibrium in imperfect-information games through self-play. One limitation of the original paper is that they used the Deep Q-Network (DQN), which approximates the best strategy based solely on expected values, creating risk-neutral agents. We want to take a step further and explore other algorithms on top of DQN that has shown promising performance recently, including distributional reinforcement learning. One algorithm which we are interested in testing is the Implicit Quantile Networks (IQN) [1], which models the full return distribution, rather than the expected values. This is interesting because the model can be made to be risk-averse or risk-seeking through customization of the agent's utility function. This allows us to model the Nash equilibrium of more realistic agents playing the game.

Project objective. Our primary goal is to reproduce the experimental setup and key findings in Heinrich and Silver[2]. Then, further algorithms (and tweaks to those algorithms), will be introduced in the original framework to observe how the result deviates from the original algorithm. For simplicity, and due to the high number of experiments we will be running, we will focus on Leduc poker (with an extension to Limit Texas Holdem if time permitted), which is a simplified version of poker with approximately 936 decision-making states. This will allow us to better assess the model's performance than the more popular Texas Hold'em, which contains up to 10^{126} decision-making states along with a very high variance payoff. Building on this, we will investigate whether introducing distributional reinforcement learning improves the Nash equilibrium convergence rate, exploitability of the learn algorithm, and improvements compared to the baseline NFSP.

Environment and implementation. We will build our implementation using the OpenSpiel research framework¹, along with any other framework we find relevant during the research process. OpenSpiel provides standardized environments for imperfect-information games, including Leduc poker, as well as exploitability evaluation tools. Using an existing environment allows us to focus on exploring algorithms more efficiently, instead of having to rebuild the game from scratch.

Expected challenges. We are expected to run into challenges of high computational cost, sensitivity of RL algorithms to hyperparameter tuning, and most importantly, the integration of distributional RL and other algorithm to existing framework. Careful experimental design would also be a crucial component to ensure a fair and statistically significant comparison between these algorithms - further review of previous studies would definitely be necessary.

Team organization. Our team has three members. We will provide space for each member to implement, test, and evaluate their algorithm independently, while sharing responsibilities in building shared architectures, writing and reporting results, as well as discussing potential improvements. This flexibility ensures the most time-consuming (and challenging) component of training the model, as well as the whole project pipeline, is efficiently studied and owned by all members. We will meet on a biweekly basis on

¹https://github.com/google-deepmind/open_spiel

Monday and Friday to go over our independent research and shared responsibilities for each milestone.

References

- [1] Dabney, W., Ostrovski, G., Silver, D., and Munos, R. (2018). Implicit quantile networks for distributional reinforcement learning. *arXiv preprint arXiv:1806.06923*.
- [2] Heinrich, J. and Silver, D. (2016). Deep reinforcement learning from self-play in imperfect-information games. *arXiv preprint arXiv:1603.01121*.